

# Band-Limitedness of the Hardy Z-Function in the Phase Variable

Stephen Crowley

April 21, 2026

## Abstract

Let  $\zeta$  denote the Riemann zeta function,  $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$  the Hardy Z-function, and  $\theta(t) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$  the Riemann–Siegel theta function. Set  $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$ ; for any constant  $c > c_*$ , define the shifted phase  $\Theta(t) := \theta(t) + ct$ . Then  $\Theta'(\tau) = \theta'(\tau) + c > 0$  for every  $\tau \in \mathbb{R}$  and  $\Theta : \mathbb{R} \rightarrow \mathbb{R}$  is a strictly increasing  $C^\infty$  bijection. The unitary inverse phase pullback

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}, \quad (1)$$

with the principal (positive) square root of the strictly positive real quantity  $\Theta'(\Theta^{-1}(u))$ , is a real-valued function on  $\mathbb{R}$  whose Fourier support is contained in  $[-1, 1]$ . By the Paley–Wiener theorem,  $Y$  extends to an entire function of exponential type  $\leq 1$ . Non-negativity of the spectral measure  $S := |\widehat{Y}|^2$  together with Akhiezer’s theorem places  $Y$  in the Laguerre–Pólya class, so all complex zeros of  $Y$  are real. On  $\mathbb{R}$  the identity  $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$  is the rearrangement of (1), with  $\sqrt{\Theta'(t)} > 0$ . This unitary time change by a strictly increasing  $C^\infty$  bijection of  $\mathbb{R}$  is a bijection between the zero set of  $Y$  on  $\mathbb{R}$  and the zero set of  $Z$  on  $\mathbb{R}$  with preservation of multiplicity. The critical-line identity  $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$  for  $t \in \mathbb{R}$ , combined with reality of zeros of  $Y$  and the unitary transfer, fixes the zeros of  $\zeta$  on the critical line as the  $\Theta^{-1}$ -image of the real zeros of  $Y$ .

## 1 Introduction

The Hardy Z-function  $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$  is real-valued on  $\mathbb{R}$ . The equivalence  $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$  for  $t \in \mathbb{R}$  ties the zeros of  $Z$  on  $\mathbb{R}$  to the zeros of  $\zeta$  on the critical line  $\{\operatorname{Re} s = \frac{1}{2}\}$  under  $s = \frac{1}{2} + it$ : the map  $t \mapsto \frac{1}{2} + it$  is a bijection  $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$ , and  $|e^{i\theta(t)}| = 1$  for  $t \in \mathbb{R}$  makes the two sides vanish simultaneously.

The Riemann–Siegel phase  $\theta$  fails to be monotone on  $[0, \infty)$ :  $\theta'$  is strictly increasing on  $[0, \infty)$  but negative on  $[0, t_c]$  and positive on  $[t_c, \infty)$ , where  $t_c \approx 6.2898$  is the unique real zero of  $\theta'$ . Monotonicity on all of  $\mathbb{R}$  is restored by the linear shift  $\Theta(t) := \theta(t) + ct$  with  $c > c_* := -\theta'(0)$ , which enforces  $\Theta' > 0$  on  $\mathbb{R}$  and makes  $\Theta : \mathbb{R} \rightarrow \mathbb{R}$  a  $C^\infty$  bijection. The unitary inverse phase pullback  $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$  is defined in terms of this real bijection.

All spectral quantities, in particular the mode thresholds  $\beta_n$ , are given in closed form as values of the digamma function  $\psi$  at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set  $\mathcal{S}(\mu)$ .

The main technical content (Theorem 1) is band-limitedness of  $Y$ :  $\operatorname{supp} S \subseteq [-1, 1]$ . Paley–Wiener then gives  $Y$  entire of exponential type  $\leq 1$ ; Akhiezer gives reality of its complex zeros. The real-line identity  $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$  is a unitary time change by a strictly increasing  $C^\infty$  bijection of  $\mathbb{R}$ , and so is a zero-set bijection with multiplicity preservation between  $Y$  on  $\mathbb{R}$  and  $Z$

on  $\mathbb{R}$  (Theorem 3). The equivalence  $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$  on  $t \in \mathbb{R}$  yields the conclusion for the zeros of  $\zeta$  on the critical line.

## 2 The Phase Function

For  $t \in \mathbb{C}$  with  $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$ ,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (2)$$

$$\theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (3)$$

$$\theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right) \quad (4)$$

For  $t \in \mathbb{R}$ ,  $\frac{1}{4} + \frac{it}{2}$  avoids  $\mathbb{Z}_{\leq 0}$  (its real part is  $\frac{1}{4}$ ), so  $\theta, \theta', \theta''$  are  $C^\infty$  on  $\mathbb{R}$ .

**Lemma 1** (Positivity of  $\theta''$ ). *For every  $t > 0$ ,  $\theta''(t) > 0$ .*

*Proof.*  $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$ . With  $z_k = \frac{1}{4} + k + \frac{it}{2}$ ,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \quad (k \geq 0, t > 0) \quad (5)$$

Summing,  $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$ , therefore  $\theta''(t) > 0$ .  $\square$

**Lemma 2** (Exact value of  $\theta'(0)$ ).

$$\theta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2} \left( \gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right) \quad (6)$$

*Proof.*  $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$  from  $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$  (reflection at  $x = \frac{1}{4}$ ) and  $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$  ([1], §6.3.4).  $\square$

**Definition 1** (Admissible constant).

$$c_* := -\theta'(0) = \frac{1}{2} \left( \gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right). \quad (7)$$

A real  $c$  is admissible if  $c > c_*$ . Throughout the paper  $c$  is admissible.

**Lemma 3** (Monotone bijection). *For every admissible  $c$ ,  $\Theta'(\tau) = \theta'(\tau) + c > 0$  for every  $\tau \in \mathbb{R}$ , and  $\Theta : \mathbb{R} \rightarrow \mathbb{R}$  is a strictly increasing  $C^\infty$  bijection.*

*Proof.*  $\theta'$  is even on  $\mathbb{R}$  (from (3);  $\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2})$  is even in  $\tau$ ) and by Lemma 1  $\theta'' > 0$  on  $(0, \infty)$ , so  $\theta'$  is strictly increasing on  $[0, \infty)$  and attains its minimum on  $\mathbb{R}$  at  $\tau = 0$ , namely  $\theta'(0) = -c_*$ . For any  $c > c_*$ ,  $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$  on  $\mathbb{R}$ . The standard asymptotic  $\theta(\tau) = \frac{\tau}{2} \log(\tau/(2\pi)) - \frac{\tau}{2} + O(1)$  as  $\tau \rightarrow \infty$  gives  $\Theta(\tau) \rightarrow +\infty$  as  $\tau \rightarrow +\infty$  and  $\Theta(\tau) \rightarrow -\infty$  as  $\tau \rightarrow -\infty$  (the second by the functional relation  $\theta(-\tau) = -\theta(\tau)$  on  $\mathbb{R}$ ). Strict monotonicity together with these limits gives that  $\Theta : \mathbb{R} \rightarrow \mathbb{R}$  is a bijection. Smoothness is inherited from  $\theta \in C^\infty(\mathbb{R})$ .  $\square$

### 3 The Unitary Inverse Phase Pullback

**Definition 2** (Unitary inverse phase pullback). *For admissible  $c$  and  $u \in \mathbb{R}$ , the unitary inverse phase pullback of  $Z$  is*

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (8)$$

*The map  $\Theta^{-1}$  is the inverse of the bijection  $\Theta : \mathbb{R} \rightarrow \mathbb{R}$  from Lemma 3;  $\Theta'(\Theta^{-1}(u))$  is a strictly positive real number, and  $\sqrt{\Theta'(\Theta^{-1}(u))}$  denotes its principal (positive) square root. The half-Jacobian factor renders the map  $Z \mapsto Y$  an  $L^2$ -isometry under the change of variable  $u = \Theta(t)$ .*

$Y$  is real-valued on  $\mathbb{R}$  because  $Z$  is real-valued on  $\mathbb{R}$  and the denominator is a strictly positive real number.

### 4 Band-Limitedness

**Definition 3** (Mode threshold). *For each  $n \geq 1$ ,*

$$\beta_n := \Theta'(2\pi n^2) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) - \frac{1}{2} \log \pi + c \quad (9)$$

The quantity  $\beta_n$  is given exactly in closed form as a value of the digamma function.

**Lemma 4** (Asymptotic of  $\beta_n$  from the exact form). *As  $n \rightarrow \infty$ ,*

$$\beta_n = \log n + c + O(n^{-4}) \quad (10)$$

*Proof.* The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k s^{2k}}, \quad |\arg s| < \pi \quad (11)$$

applied at  $s = \frac{1}{4} + i\pi n^2$  with  $|s| = \sqrt{\frac{1}{16} + \pi^2 n^4}$  yields

$$\operatorname{Re} \psi(s) = \log |s| + O(|s|^{-2}) = \frac{1}{2} \log\left(\frac{1}{16} + \pi^2 n^4\right) + O(n^{-4}) = 2 \log n + \log \pi + O(n^{-4}) \quad (12)$$

Therefore  $\frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) = \log n + \frac{1}{2} \log \pi + O(n^{-4})$  and  $\beta_n = \log n + c + O(n^{-4})$ .  $\square$

**Definition 4** (Truncated transform). *For  $T > 0$ ,  $\mu \in \mathbb{R}$ ,*

$$K_T(\mu) := \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du \quad (13)$$

**Lemma 5** (Riemann–Siegel identity). *For  $t > 0$ ,*

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + R(t), \quad N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}) \quad (14)$$

*Proof.* [3], §7.4, Theorem 7.1.  $\square$

**Theorem 1** (Band-limitedness of  $Y$ ). *For every admissible  $c$  and every  $\mu \in \mathbb{R}$  with  $|\mu| > 1$ ,  $\lim_{T \rightarrow \infty} K_T(\mu) = 0$ . Equivalently,  $\operatorname{supp} S \subseteq [-1, 1]$ , where  $S := |\widehat{Y}|^2$ .*

*Proof.* Substituting Lemma 5 into  $K_T$  and expanding  $\cos \alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$ ,

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n \leq N(T)} n^{-1/2} J_{n,\sigma}(T, \mu) + \mathcal{R}(T, \mu) \quad (15)$$

with  $t_n := 2\pi n^2$ ,

$$J_{n,\sigma}(T, \mu) := \int_{t_n}^T \sqrt{\Theta'(t)} e^{i[\sigma\theta(t) - \sigma t \log n - \mu\Theta(t)]} dt \quad (16)$$

$$\mathcal{R}(T, \mu) := \frac{1}{2\pi} \int_0^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt \quad (17)$$

The argument proceeds in the following steps.

(i) *Remainder vanishing.*  $R(t)\sqrt{\Theta'(t)} = O(t^{-1/4}(\log t)^{1/2})$ ; the phase derivative  $|\mu\Theta'(t)| \geq |\mu|(c - c_*)$  and grows as  $\frac{|\mu|}{2} \log(t/(2\pi))$ . Integration by parts gives  $\mathcal{R}(\infty, \mu) - \mathcal{R}(T, \mu) = O(T^{-1/4}(\log T)^{-1/2}) \rightarrow 0$ .

(ii) *Change of variable.* Set  $x := \Theta'(t) = \theta'(t) + c$ ,  $dx = \theta''(t) dt$ ,  $X := \Theta'(T)$ :

$$J_{n,\sigma}(T, \mu) = \int_{\beta_n \vee \Theta'(0)}^X \frac{\sqrt{x}}{\theta''(t(x))} e^{i\tilde{\Phi}_{n,\sigma,\mu}(x)} dx \quad (18)$$

$$\tilde{\Phi}'_{n,\sigma,\mu}(x) = \frac{(\sigma - \mu)x - \sigma \log n - \sigma c}{\theta''(t(x))} \quad (19)$$

(iii) *Exact  $\theta''$  cancellation.*

$$Q_{n,\sigma,\mu}(x) = \frac{\sqrt{x}}{(\sigma - \mu)x - \sigma \log n - \sigma c}, \quad Q'_{n,\sigma,\mu}(x) = O(x^{-3/2}) \quad (20)$$

(iv) *Finiteness of the exceptional set.* Define  $\mathcal{S}(\mu) := \{n : \beta_n \leq (\log n + c)/(1 + |\mu|)\}$ . By Lemma 4,

$$\beta_n - \frac{\log n + c}{1 + |\mu|} = \frac{|\mu|}{1 + |\mu|} (\log n + c) + O(n^{-4}) \rightarrow \infty \quad (21)$$

therefore  $\mathcal{S}(\mu)$  is finite.

(v) *IBP for  $n \notin \mathcal{S}(\mu)$ .* Constant-sign phase derivative; boundary  $O(X^{-1/2})$ , tail  $\leq 2CX^{-1/2}$ .

(vi) *IBP tail for  $n \in \mathcal{S}(\mu)$ .* Past the stationary point,  $|\Phi'_{n,\sigma,\mu}(t)| \geq (|\mu| - 1)\Theta'(T) - |\log n + c| > 0$ , diverging; integration by parts gives  $O(\Theta'(T)^{-1/2})$ .

(vii) *Assembly.*

$$K_\infty(\mu) - K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma} \left[ \sum_{n \notin \mathcal{S}(\mu)} n^{-1/2} O(X^{-1/2}) + \sum_{n \in \mathcal{S}(\mu)} n^{-1/2} O(\Theta'(T)^{-1/2}) \right] + O(T^{-1/4}(\log T)^{-1/2}) \quad (22)$$

All terms vanish as  $T \rightarrow \infty$ ;  $K_\infty(\mu)$  exists.

(viii) *Identification.* The mode  $(n, \sigma)$  instantaneous  $u$ -frequency is

$$\sigma \cdot \frac{\theta'(t) - \log n}{\Theta'(t)} = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c} \quad (23)$$

On  $t \geq 2\pi n^2$ ,  $\theta'(t) \geq \theta'(2\pi n^2) = \beta_n - c = \frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + i\pi n^2) - \frac{1}{2} \log \pi$ . Numerator  $\geq 0$ ; denominator exceeds numerator by  $c + \log n > 0$ ; ratio in  $[0, 1)$ . The Riemann–Siegel remainder has  $u$ -frequency  $O((\log t)^{-1}) \rightarrow 0$ .

Therefore  $\operatorname{supp} S \subseteq [-1, 1]$ , and  $K_\infty(\mu) = \widehat{Y}(\mu) = 0$  for  $|\mu| > 1$ .  $\square$

## 5 Entire Extension and Laguerre–Pólya Membership

**Corollary 1** (Paley–Wiener).  *$Y$  extends to an entire function of exponential type  $\leq 1$ :*

$$Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi, \quad |Y(z)| \leq \|\widehat{Y}\| e^{|\operatorname{Im} z|} \quad (24)$$

*Proof.*  $Y$  is a tempered distribution on  $\mathbb{R}$  whose Fourier transform is supported in the compact set  $[-1, 1]$ . By [4], Theorem 7.3.1 (distributional Paley–Wiener),  $Y$  extends to an entire function of exponential type with type constant bounded by the support radius, which here is 1; the representation  $Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi$  follows from the same theorem, and  $|Y(z)| \leq \|\widehat{Y}\| e^{|\operatorname{Im} z|}$  is the bound on the right.  $\square$

**Lemma 6** (Non-negativity of the spectral measure). *The spectral measure  $S := |\widehat{Y}|^2$  is a non-negative tempered measure on  $\mathbb{R}$ , supported in  $[-1, 1]$ .*

*Proof.* The autocorrelation  $C(\eta) := \langle Y, Y(\cdot + \eta) \rangle$  is positive-definite in the tempered sense; Bochner–Schwartz ([7], Theorem 7.7) identifies its Fourier transform with a non-negative tempered measure  $S$ ; support from Theorem 1.  $\square$

**Theorem 2** (Akhiezer). *A real-valued entire function  $F$  of exponential type  $\leq \tau$  whose spectral measure  $|\widehat{F}|^2$  is a non-negative measure supported in  $[-\tau, \tau]$  belongs to the Laguerre–Pólya class. All zeros of  $F$  in  $\mathbb{C}$  are real.*

*Proof.* [5], Lecture 16, Theorem 3; [2], §V.4.  $\square$

**Corollary 2** (Reality of zeros of  $Y$ ). *All zeros of  $Y$  in  $\mathbb{C}$  are real.*

*Proof.*  $Y$  is real on  $\mathbb{R}$  (Definition 2), entire of exponential type  $\leq 1$  (Corollary 1), with spectral measure  $S$  non-negative and supported in  $[-1, 1]$  (Lemma 6). Apply Theorem 2.  $\square$

## 6 Transfer to the Hardy Z-Function on the Real Line

The transfer uses only the real-line identity

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)}, \quad t \in \mathbb{R}, \quad (25)$$

which is the rearrangement of (8) after substituting  $u = \Theta(t)$ . On  $\mathbb{R}$ ,  $\Theta'(t) > 0$  (Lemma 3) and  $\sqrt{\Theta'(t)}$  is the principal (positive) square root of a strictly positive real number; no branch choice is involved.

**Theorem 3** (Zero-set bijection under the unitary time change on  $\mathbb{R}$ ). *Let  $c$  be admissible. The map  $t \mapsto \Theta(t)$  is a bijection  $\mathbb{R} \rightarrow \mathbb{R}$  that restricts to a bijection of zero sets,*

$$\{t \in \mathbb{R} : Z(t) = 0\} \longleftrightarrow \{u \in \mathbb{R} : Y(u) = 0\}, \quad u = \Theta(t), \quad (26)$$

*with preservation of multiplicity: if  $t_0 \in \mathbb{R}$  and  $Y$  has a zero of order  $m$  at  $\Theta(t_0)$ , then  $Z$  has a zero of order exactly  $m$  at  $t_0$ , and conversely.*

*Proof.*  $\sqrt{\Theta'(t)}$  is strictly positive on  $\mathbb{R}$ . From (25), for any  $t_0 \in \mathbb{R}$ ,

$$Z(t_0) = 0 \iff \sqrt{\Theta'(t_0)} Y(\Theta(t_0)) = 0 \iff Y(\Theta(t_0)) = 0. \quad (27)$$

Since  $\Theta : \mathbb{R} \rightarrow \mathbb{R}$  is a bijection (Lemma 3),  $t_0 \mapsto \Theta(t_0)$  is a bijection between the real zero set of  $Z$  and the real zero set of  $Y$ .

For multiplicity, let  $t_0 \in \mathbb{R}$  and suppose  $Y$  has a zero of order  $m$  at  $s_0 := \Theta(t_0)$ . Since  $Y$  is entire (Corollary 1), there exists an entire  $g$  with  $g(s_0) \neq 0$  and

$$Y(s) = (s - s_0)^m g(s). \quad (28)$$

Substitute  $s = \Theta(t)$  (defined and  $C^\infty$  for  $t \in \mathbb{R}$ ):

$$Y(\Theta(t)) = (\Theta(t) - \Theta(t_0))^m g(\Theta(t)). \quad (29)$$

The Hadamard-type identity

$$\Theta(t) - \Theta(t_0) = (t - t_0) h(t), \quad h(t) := \int_0^1 \Theta'(t_0 + u(t - t_0)) du, \quad (30)$$

valid for all  $t \in \mathbb{R}$  because  $\Theta \in C^1(\mathbb{R})$ , gives  $h(t_0) = \Theta'(t_0) > 0$ , and continuity of  $h$  together with  $\Theta' > 0$  on  $\mathbb{R}$  ensures  $h(t) > 0$  for  $t$  in a neighborhood of  $t_0$ . Combining (25), (29), and (30),

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)} = (t - t_0)^m h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (31)$$

Denote the trailing factor

$$H(t) := h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (32)$$

Then  $H$  is continuous at  $t_0$  and

$$H(t_0) = \Theta'(t_0)^m g(s_0) \sqrt{\Theta'(t_0)} \neq 0 \quad (33)$$

because  $\Theta'(t_0) > 0$  and  $g(s_0) \neq 0$ . Therefore  $Z$  has a zero of order exactly  $m$  at  $t_0$ .

The converse (given a zero of  $Z$  of order  $m$  at  $t_0 \in \mathbb{R}$ ,  $Y$  has a zero of order  $m$  at  $\Theta(t_0)$ ) is obtained by applying the same argument with  $\Theta$  replaced by  $\Theta^{-1}$ , which satisfies  $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$  for  $u \in \mathbb{R}$  by Lemma 3 and the inverse function theorem.  $\square$

**Corollary 3** (Zeros of  $Z$  on  $\mathbb{R}$ ). *The zero set of  $Z$  on  $\mathbb{R}$ , counted with multiplicity, is  $\Theta^{-1}$  applied to the zero set of  $Y$ , counted with multiplicity.*

*Proof.* Corollary 2 gives that every zero of the entire function  $Y$  lies on  $\mathbb{R}$ . Theorem 3 gives a multiplicity-preserving bijection between the zero set of  $Y$  on  $\mathbb{R}$  and the zero set of  $Z$  on  $\mathbb{R}$ , implemented by  $t \mapsto \Theta(t)$ .  $\square$

## 7 The Zero Locus of $\zeta$ on the Critical Line

**Corollary 4** (Zeros of  $\zeta$  on the critical line). *The zero set of  $\zeta$  on the critical line  $\{\operatorname{Re} s = \frac{1}{2}\}$ , counted with multiplicity, is the image of  $\Theta^{-1}(\{u \in \mathbb{R} : Y(u) = 0\})$  under the map  $t \mapsto \frac{1}{2} + it$ , with multiplicities inherited from those of  $Y$ .*

*Proof.* For  $t \in \mathbb{R}$ ,  $|e^{i\theta(t)}| = 1$ , so  $e^{i\theta(t)}$  is a nonzero complex number; hence  $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$  gives  $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ , with matching orders of vanishing at every  $t \in \mathbb{R}$  (multiplication by the non-vanishing real-analytic factor  $e^{-i\theta(t)}$  does not alter the order of a zero of the real-analytic factor  $Z(t)$  at a real point). Corollary 3 identifies the real zero set of  $Z$  with  $\Theta^{-1}$  applied to the zero set of  $Y$ . The map  $t \mapsto \frac{1}{2} + it$  is a bijection  $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$ . Composing gives the stated identification.  $\square$

## References

- [1] M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions*, Dover, 1972.
- [2] N. I. Akhiezer, *Theory of Approximation*, Ungar, 1956.
- [3] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.
- [4] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, 2nd ed., Springer, 1990.
- [5] B. Ya. Levin, *Lectures on Entire Functions*, Translations of Mathematical Monographs, vol. 150, American Mathematical Society, 1996.
- [6] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.
- [7] W. Rudin, *Functional Analysis*, 2nd ed., McGraw–Hill, 1991.
- [8] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.